CricketMind

Verified Claims over Real Evidence

Thesis Architecture Draft — Version 3

Status. This is a design. No v3 component has been built or measured. Every number in this document is a design parameter, a figure cited from the literature, or an illustrative computation on the synthetic corpus of the v2 reference implementation, and is labelled as such. The accuracy of the perception front end is evaluated separately and is not reported here.

1 What Changes in Version 3, and Why

Version 2 established the control flow: two entry paths, two bounded correction loops, twelve stakeholder agents, and a Faith Check that compares every cited figure against two sanctioned engines. Its evaluation also exposed four gaps, and Version 3 is organised around closing them.

1. **The gate verifies values, not claims.** It rejected fabricated citations but passed a correct figure attached to the wrong subject and a figure stated without a citation. The conclusion of that evaluation already names a subject-keyed citation contract as a requirement for language-model agents.

2. **Four agents assert claims that no engine computes.** The Stadium agent says the surface "is holding pace" while citing only a ball count; the Compliance agent reports "no eligibility flags" with no compliance data; the Injury agent says a workload "sits inside the weekly threshold" without computing a workload, and counts balls *faced*; the Sponsorship agent names the "most replayed shot" with no replay data. Each passes the gate, because the number it cites is correct. Three more agents make salary, valuation or selection recommendations with no market data behind them.

3. **Predictive claims are verified as if they were facts.** N9 already computes a 95% interval, but no agent cites it and the gate never reads it. Worse, that interval measures only Monte Carlo error (Section 4.6).

4. **The evidence is synthetic, and real ball-by-ball data has no shot type.** Cricsheet records every delivery but not which stroke was played, so the shot the video identifies has nowhere to land.

Table 1 lists the changes. The principle of v2 is kept and sharpened: numbers come only from sanctioned sources, and now every *kind of claim* must come from one too.

#	Change	Motivation	Where
1	Subject-keyed claim contract	Gate blind to subject swaps and uncited figures (gap 1)	All agents; Faith Check C1, C2, C5, C6
2	Uncertainty-faithful verification	Interval computed but unused; Monte Carlo error only (gap 3)	N9; Faith Check C3, C4
3	Engines behind hollow agents, and a claim-type registry	Claims with no engine behind them (gap 2)	N8 modules M2–M4; registry; N11f, N11i, N11k
4	Real evidence, and N9 Tier 2 matchups with calibration	Synthetic corpus (gap 4)	Delivery DB; N7a, N7b; N8 M1; N9 T2
5	Commentary-to-shot bridge	No shot type in ball-by-ball data (gap 4)	Offline pipeline O3
6	Measured posture from pose estimation	Posture was a per-shot lookup constant	N2b; N6
R	Roster: Bowling Performance added; Compliance → Match Officials; Sponsorship merged into Content Production	No bowler-side agent; no public data can back compliance or sponsorship claims	N11g, N11h, N11i

Table 1: Changes from Version 2. Node letters are kept, so code identifiers map one-to-one; N11g and N11i change role.

2 Design Invariants

Version 3 is defined by six invariants. Every later section is an implementation of one of them.

I1 — Numeric authority.
Only three sources may produce a number: N6 (*measured* from the clip), N8 (*descriptive*, counted from history) and N9 (*predictive*, simulated). There are still two engines; N6 is a measurement, as it already was in v2. N9 remains code and never becomes a language-model call.

I2 — No hollow claims.
Every claim type an agent may assert resolves to a metric of one sanctioned source. This is checked when the graph is built, so a template that asserts an unsupported kind of claim cannot be deployed.

I3 — Agents never type numbers.
Agents write prose that references claim identifiers. A deterministic renderer inserts the verified values.

I4 — Uncertainty travels with the claim.
Predictive and comparative claims carry their interval, and their wording must be consistent with it.

I5 — No measurement without a clip.
A text-only query can never carry a measured claim; the posture fields are absent, not defaulted.

I6 — Bounded cost.
Every loop is bounded (at most two retries, two regenerations and four active agents), so worst-case cost stays closed-form.

3 System Architecture Diagram

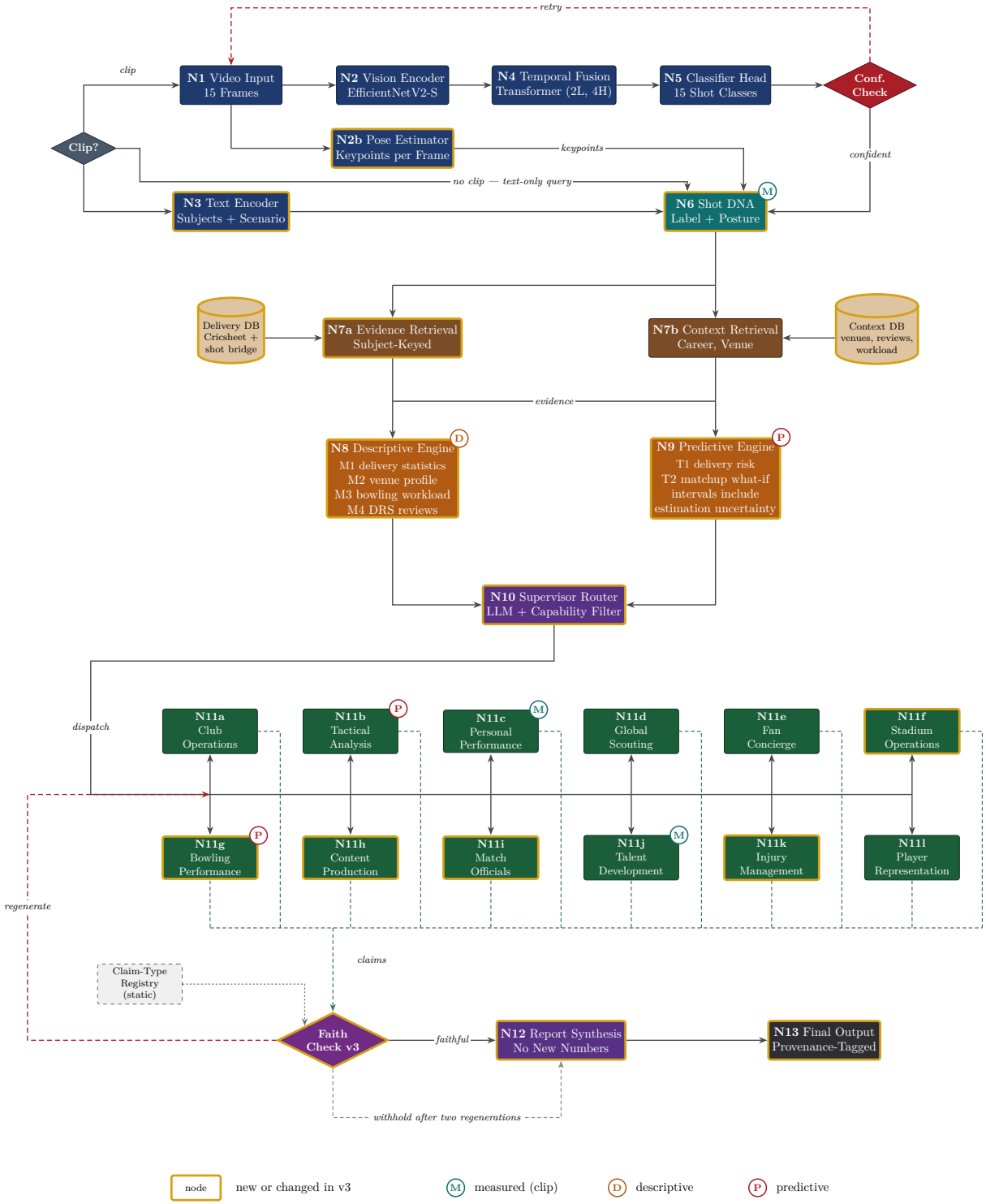


Figure 1: CricketMind LangGraph architecture, Version 3. Gold outlines mark nodes that are new or changed since v2. Badges give the provenance class a node produces or an agent may cite: M for measured from the clip, D for descriptive (counted from history), P for predictive (simulated). Agents emit claim tuples rather than numbers; the Faith Check re-queries the engines for each tuple (checks C1–C6, Figure 3) against the static claim-type registry. Failing agents are regenerated with the reasons for failure, at most twice, after which the finding is withheld and listed in the report. The pose estimator N2b reads the same frames as N2 and runs in parallel with the encoder chain. Data sources and the offline commentary-to-shot bridge are shown in Figure 2.

4 Node-by-Node Changes

N1, N2, N4, N5 and the Confidence Check are unchanged from v2: fifteen frames, an EfficientNetV2-S frame encoder, a two-layer temporal transformer and a fifteen-class head, with at most two retries. The vision backbone remains a swappable component, and its selection is reported in a separate study.

4.1 N2b Pose Estimator and Measured Posture (Change 6)

In v2, N6 looked posture up in a table keyed by the predicted shot: every cover drive received the same weight transfer. N2b replaces the lookup with a measurement. A pose model [5] extracts seventeen body keypoints per frame on the same fifteen frames, reusing the striker selection of the CricShot10k preprocessing. Three posture metrics are computed from the keypoints:

- **Weight transfer**: the position of the hip midpoint between the back and front ankles at the frame of peak wrist speed, scaled to $[0, 1]$. The front foot is the ankle that moves further during the clip.
- **Stride length**: the largest ankle separation, divided by leg length (hip to ankle), so it does not depend on camera distance.
- **Head over front knee**: the horizontal offset between head and front knee at peak wrist speed, divided by leg length.

The v2 field `bat_face_angle` is dropped: body keypoints contain no bat.

Validation without new labels. Posture has no ground truth, but the CricShot10k labels supply a proxy. Shots played predominantly off the front foot (Cover Drive, Straight Drive, Sweep, Reverse Sweep, Down The Wicket) should show higher weight transfer than shots played predominantly off the back foot (Pull, Hook, Square Cut, Late Cut, Upper Cut); ambiguous classes are excluded. The test is the ROC-AUC of each metric in separating the two groups on the test split, with an acceptance threshold of 0.75 (a design parameter). A metric that fails is removed from the claim-type registry, and agents simply cannot cite it. This test uses dataset labels, not the classifier, and reports no classifier accuracy.

4.2 N3 Text Encoder: Subjects and Scenario

N3 now emits a structured parse under a fixed schema: batter, bowler(s), shots, phase, bowler type, venue, format, and an optional scenario (*delivery* with a ball count k, or *matchup* with two bowlers). Every name is resolved against the player register. A player with no deliveries in the evidence is reported as such, closing a gap the v2 evaluation listed: the router then wakes no agent whose claims would need that player's data.

4.3 N6 Shot DNA

N6 fuses the accepted shot label, the measured posture (provenance M, video path only, invariant I5) and the parsed subjects into one record. Downstream nodes still never see a pixel.

4.4 Evidence Layer and Retrieval (Changes 4 and 5)

Source	Provides	Does not provide	Access and caveats
CricShot10k [1]	15-class batting-shot clips	identity, outcome	held locally
Cricsheet [2]	every delivery: batter, bowler, runs, extras, wicket, venue, date; DRS reviews attached to the delivery	shot type, line and length, bowler type	free with attribution; review data has needed corrections [3]
Player register with Cricinfo attributes [4]	batting hand, bowling type	—	identifiers link the two; check terms of use
Commentary corpus	text that names the stroke played	shot labels; match and ball keys unverified	must be licensed; schema verified before O3
Injury, replay, sponsorship, eligibility records	nothing public	—	claims needing them are not registered

Table 2: Evidence sources. The last row is why the Compliance and Sponsorship roles cannot be backed by any engine.

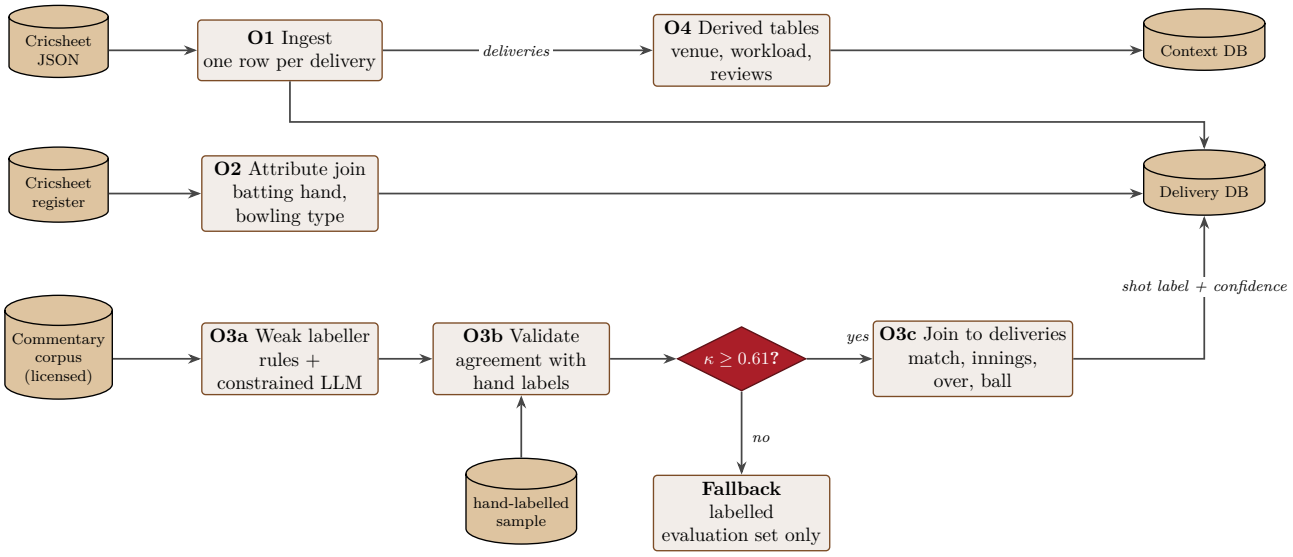


Figure 2: Offline evidence pipeline. Nothing here runs at query time. The commentary bridge (O3) is accepted only if its labels agree substantially with a hand-labelled sample; otherwise shot-keyed statistics are restricted to a hand-labelled evaluation set.

O1, O2, O4: ball-by-ball evidence. O1 flattens Cricsheet into one row per delivery. O2 attaches batting hand and bowling type through the player register, so that filters such as "against pace" have a source. O4 derives the tables the new N8 modules read: venue aggregates, balls bowled per bowler per date, and the DRS reviews attached to deliveries.

O3: the commentary-to-shot bridge. Ball-by-ball commentary names the stroke ("drives through cover", "pulls to deep square"), but no labelled dataset pairs commentary with the fifteen CricShot10k classes. O3a labels each line weakly [6]: keyword rules first, and for lines the rules cannot resolve, a language model whose output is constrained to the fifteen classes or *none*. O3b measures agreement with a stratified hand-labelled sample of about 500 lines by Cohen's κ [7] and per-class F1; the bridge is accepted at $\kappa \geq 0.61$, "substantial" agreement [8]. O3c joins labels to deliveries by match, innings, over and ball. Wides and no-balls make this join error-prone, as Cricsheet's own corrections to its review data show [3], so each label carries its confidence and origin (rule or model). The language model is used offline, to produce categories that are validated before use; it produces no number an agent can cite, so invariant I1 is untouched.

N7a and N7b. N7a filters by the full subject tuple from N3 (batter, bowler, shot, phase, bowler type, venue) and returns the matching deliveries with their count. A shot filter applies only to deliveries the bridge labelled. N7b retrieves career and venue context as before.

4.5 N8 Descriptive Engine (Changes 3 and 4)

N8 stays deterministic aggregation: every output can be recomputed by hand. It gains three modules, each of which backs an agent that had no engine before.

M1 — delivery statistics.
Strike rate, dismissal rate and sample size for any subject tuple, plus differences between two tuples with a bootstrap 95% interval (used by check C4).

M2 — venue profile
(backs N11f). Average first-innings score, share of matches won chasing, and the share of wickets taken by pace and by spin.

M3 — bowling workload
(backs N11k). Balls bowled in the last 7 and 28 days and their ratio, the acute:chronic workload ratio. Workload spikes were associated with roughly double the injury risk in the following week in elite fast bowlers [9], but the ratio's validity is disputed [10], and match data omits training loads. M3 therefore emits a *monitoring flag*, never an injury probability.

M4 — DRS reviews
(backs N11i). Review counts and overturn rates by team and by dismissal type.

4.6 N9 Predictive Engine and Honest Uncertainty (Changes 2 and 4)

T1 and T2. Tier 1 is v2's delivery question ("six short balls: what is the chance of a wicket?"). Tier 2 adds matchup counterfactuals ("what if this bowler bowled the over instead of that one?"), estimated from head-to-head rates shrunk toward population rates [11]. Following the N9 design note, a counterfactual is always reported as a difference against a baseline, with an interval; if the interval contains zero, the output is *no detectable difference*. Tier 3 (batting order, chases) remains future work.

Two kinds of uncertainty. v2's interval measures only Monte Carlo error, which shrinks as trials increase. It omits estimation uncertainty, which shrinks only with more data. The difference is large. On the synthetic corpus (illustrative only), the batter's pull has 4 dismissals in 50 balls:

	P(out within 6 balls)	**95% interval**
v2: Monte Carlo error only (20,000 trials)	0.394	[0.387, 0.400]
v3: estimation uncertainty propagated	0.405 (median)	[0.155, 0.697]

The v3 row draws the per-ball rate from a Beta posterior (Jeffreys prior; shrinkage toward the population would change it again) and pushes each draw through the simulation. The honest interval is about twenty times wider, and it spans three wording bands of Table 5. v2's interval would have licensed a confidence the evidence does not support.

Calibration. T1 and T2 are calibrated on a held-out season: predicted probabilities against observed frequencies (reliability diagrams) and the Brier score [12]. An optional adjustment of rates by measured posture stays off by default until it is validated, as the N9 design note requires.

4.7 N10 Supervisor Router

The language-model router now receives a short evidence summary: which claim types are satisfiable for this query (posture only with a clip, matchups only with two named bowlers, venue claims only with a venue). Fixed rules then apply a capability filter (an agent with no satisfiable claim type is not woken), the cap of four active agents, and the default pair. The bounds of v2 are unchanged.

4.8 The Twelve Stakeholder Agents

Stakeholder	Agent	Claim types it may assert	Prov.	Versus v2
Franchises / Boards	Club Operations (N11a)	strike rate, dismissal rate, sample size; venue profile	D	no salary claims without a market engine
Coaches	Tactical Analysis (N11b)	strike and dismissal rates; P(dismissal within k); matchup difference	D, P	interval now mandatory
Players	Personal Performance (N11c)	strike rates; posture metrics	D, M	posture now measured
Scouts	Global Scouting (N11d)	strike rate, career strike rate, league benchmark	D	—
Fans	Fan Concierge (N11e)	venue strike rate, strike rate, recent form	D	advice must cite D facts
Stadiums / curators	Stadium Operations (N11f)	first-innings average, chasing share, pace and spin wicket share	D	now backed by M2
Bowlers / bowling coaches	**Bowling Performance (N11g)**	economy and dismissal rate by phase and batting hand; matchup difference	D, P	**new**; replaces Sponsorship
Broadcasters / sponsors	Content Production (N11h)	strike rates; boundary counts as a visibility proxy	D	absorbs Sponsorship
Governing bodies / umpires	**Match Officials (N11i)**	review counts, overturn rate by team and dismissal type	D	**reframed** from Compliance; backed by M4
Academies	Talent Development (N11j)	strike rates; posture metrics	D, M	posture now measured
Medical teams	Injury Management (N11k)	balls bowled (7 and 28 days), workload ratio flag	D	backed by M3; a flag, never a prediction
Player agents	Player Representation (N11l)	strike rate, career strike rate	D	no valuation claims without a market engine

Table 3: The Version 3 roster. Every claim type in the third column is a registry entry backed by an engine metric (invariant I2). Adding a market engine from public auction prices would restore valuation claims for N11a and N11l: one registry entry and one N8 module, with no change to the graph.

Two agents now cite predictions (N11b, N11g); in v2 only N11b did. Every current agent reasons about the batter, so N11g fills a real coverage gap; Cricsheet records the bowler of every delivery.

5 The Claim Contract (Change 1)

An agent finding is a list of claims and a prose template. The claim is the unit of verification:

```
{"id": "c1", "kind": "descriptive", "metric": "strike_rate", "engine": "N8.M1",
 "subject": {"player": "Kohli", "shot": "Cover Drive", "phase": "powerplay",
             "bowler_type": "pace"},
 "value": 142.0, "n": 50}
{"id": "c2", "kind": "predictive", "metric": "p_dismissal_within_k", "engine": "N9.T1",
 "subject": {"player": "Kohli", "shot": "Pull", "phase": "powerplay",
             "bowler_type": "pace", "k": 6},
 "value": 0.405, "ci95": [0.155, 0.697], "band": "about even"}

template: "Against pace in the powerplay, Kohli's cover drive returns {c1}.
           Six short balls carry {c2} of a dismissal."
```

A deterministic renderer replaces each reference with the verified value and its provenance tag, for example "a strike rate of 142.0 [D, 50 balls]". The agent never types a digit (invariant I3). Findings are produced under constrained decoding against this schema; a finding that does not parse fails check C1.

The claim-type registry. Table 4 shows part of the registry. It is built with the graph, and two things are checked then: every claim type an agent declares exists, and every registry entry resolves to an engine metric. A template that asserts "the surface is holding pace" with no venue metric behind it is therefore not a runtime failure but a build failure.

Claim type	Prov.	Source	Required subject keys	Checks
`strike_rate`	D	N8.M1	player; shot, phase, bowler type optional	C2 exact; C3 floor
`dismissal_rate`	D	N8.M1	as above	C2; C3
`sr_difference`	D	N8.M1	player, two shots; phase, bowler type optional	C2; C4 zero-crossing
`venue_first_innings_avg`	D	N8.M2	venue, format	C2; C3 (matches)
`workload_ratio_flag`	D	N8.M3	bowler, date	C2; wording: flag only
`review_overturn_rate`	D	N8.M4	team or dismissal type	C2; C3
`weight_transfer`	M	N6	clip	C2; video path only
`p_dismissal_within_k`	P	N9.T1	player, shot, phase, bowler type, k	C2 with interval; C4 band
`matchup_difference`	P	N9.T2	batter, two bowlers, phase, k	C2 with interval; C4 zero-crossing

Table 4: Part of the claim-type registry.

6 Faith Check v3

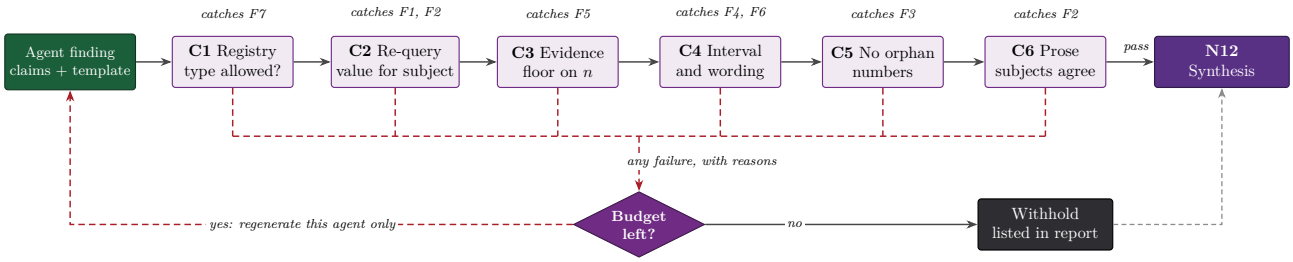


Figure 3: Faith Check v3. The six checks run in order, cheapest first, inside the single Faith Check node. Each catches particular fabrication classes of Table 6.

C1 — Schema and registry.
The finding parses; each claim type exists and this agent is allowed to assert it; the claim names its source.

C2 — Subject-bound value.
The gate re-queries the named source with the claim's subject tuple and compares the result within

rounding, including the interval for predictive claims. A correct number cited for the wrong subject fails here, because the engine returns a different value for that subject.

C3 — Evidence floor.
A descriptive claim on fewer than $n_{\min}$ balls (a design parameter; default 30) must carry a small-sample qualifier, and may not be generalised ("struggles against spin") without one.

C4 — Interval and wording.
A predictive claim's wording must fall in the band of Table 5 that contains its point estimate; if the interval crosses a band boundary, a hedge naming the evidence is required. A comparative claim, predictive or descriptive, whose difference interval contains zero must say *no detectable difference*.

C5 — No orphan numbers.
Apart from claim references and a small allowlist ("T20", "U-19"), the template contains no digits. This closes the uncited-figure gap of v2 by construction.

C6 — Prose subjects agree.
Using a gazetteer built from the player register, the fifteen shot names and their synonyms, phases, bowler types and venues, every entity named in a sentence that contains a claim reference must agree with that claim's subject.

Worked example: the v2 running example. On the synthetic corpus the batter's cover drive returns a strike rate of 142.0 and his pull 112.0, both on 50 balls. The bootstrap 95% interval for the difference is $[-16.0,\ +76.0]$, which contains zero. Under C4, the v2 template's verdict that the cover drive is "the stronger stroke" is a spurious difference (class F6): the agent must report no detectable difference on this evidence. The example is illustrative, but it shows that the stricter gate changes conclusions, not only formatting.

Point estimate p	**Required wording (any of)**	**Forbidden in this band**
$p \le 0.13$	almost certainly not, remote	likely, probable, possible
$0.13 < p \le 0.40$	probably not, unlikely	likely, probable, expected
$0.40 < p < 0.60$	about even, as likely as not	likely, unlikely
$0.60 \le p < 0.87$	probable, likely	almost certain, will
$p \ge 0.87$	almost certain, very likely	certain, guaranteed
any band	—	will, certain, guaranteed, always, never

Table 5: Wording bands for predictive claims, after Kent's words of estimative probability [13], with his ranges closed into contiguous intervals. If the 95% interval crosses a boundary, a hedge naming the evidence ("on 4 dismissals in 50 balls") is also required.

On the Tier 1 example of Section 4.6, the median of 0.405 falls in *about even* and the interval crosses two boundaries, so an acceptable rendering is "about as likely as not, though on 4 dismissals in 50 balls the estimate is very uncertain". "Will probably get him out" fails C4.

N12 and N13. N12 composes the report from verified findings only and is held to the same rule: it references claims and types no numbers, and its output passes C5 and C6. If it fails, the report falls back to the verified findings in fixed order, which keeps the step bounded. N13 shows every number with its provenance tag and lists withheld findings.

7 Control Loops and Bounds

The two loops of v2 are unchanged in structure. The retry loop guards against perception errors; the regenerate loop now re-runs only the failing agents and passes them the failing checks and reasons. Agent invocations remain bounded by $4 \times (1 + 2) = 12$, independent of roster size. N2b runs in parallel with the N2–N5 chain, and checks C1–C6 run inside the single Faith Check node, so the superstep bound of v2 is expected to hold; it must be re-derived from the closed-form cost model and re-measured before it is claimed.

8 Evaluation Plan

Fabrication containment. The existing injection harness is extended from three fabrication classes to seven (Table 6). The v2 column follows from the v2 code, which compares cited values only; it is a prediction to be confirmed, not a measurement. Each class is measured by catch rate, and correct findings by false-rejection rate, regeneration count and report completeness.

Class	Injected error	Example	v2 gate	v3 check
F1	wrong value	strike rate 162 instead of 142	caught	C2
F2	right value, wrong subject	one batter's 142 attributed to another	missed	C2, C6
F3	uncited number	"about 150" typed in the prose	missed	C5
F4	certainty inflation	"will likely be dismissed" at $p = 0.405$	missed	C4
F5	small-sample generalisation	"struggles against spin" on six balls	missed	C3
F6	spurious difference	"the stronger stroke" when the interval contains zero	missed	C4
F7	unsupported claim type	"the surface is holding pace" with no venue metric	missed	C1, registry

Table 6: Fabrication classes for the injection harness.

Engines and evidence. N8 modules are checked by recomputation against hand-computed cases. N9 T1 and T2 are checked for calibration on a held-out season. The commentary bridge reports κ and per-class F1 against hand labels. Posture reports the front-foot versus back-foot AUC of each metric.

Routing and cost. The labelled routing set is extended with queries for N11g and N11i, since every added role is a new way for the router to be wrong. The cost model's bounds are re-derived for the v3 graph before they are measured.

9 Build Plan

Phase	Work	Days	Depends on
1 Contract	claim schema, renderer, registry; checks C1, C2, C5, C6	3–5	—
1 Contract	wording bands; checks C3, C4; N9 intervals with estimation uncertainty	3–4	—
2 Honest agents	N8 modules M2–M4; roster changes including N11g	3–5	phase 1
3 Real evidence	O1, O2, O4; M1 on Cricsheet data	3–4	—
3 Real evidence	N9 Tier 2 with calibration on a held-out season	4–5	O1, O2
Evaluation	fabrication classes F1–F7; v2 against v3	2	phases 1–2
Core total		**18–25**	
4 Optional	commentary bridge O3, with validation	10–15	O1; a licensed corpus
4 Optional	N2b pose estimator and posture validation	3–5	—

Table 7: Build plan. Estimates, not commitments. Phase 1 replaces the 1–2 day structured-citation block of the completion estimate, and phase 3 replaces its 2–3 day data-layer block.

10 Risks and Fallbacks

Risk	Effect	Fallback
No licensed commentary corpus with match and ball keys	O3 impossible	hand-label an evaluation set; shot-keyed claims only on it
Bridge labels too noisy ($\kappa < 0.61$)	wrong shot statistics	disable the shot filter on real data; report the bridge as a negative result
A 7B model breaks the claim schema	findings do not parse	constrained decoding; parse failures count as C1 rejections and regenerate
Pose fails on low-resolution clips	no posture	metrics failing validation leave the registry; agents degrade to D claims
Workload ratio validity disputed	misleading medical advice	worded as a monitoring flag; never a probability
Bowling type missing for some players	filters silently shrink n	n is always reported; C3 floor
New roles confuse the router	wrong agents woken	larger labelled routing set; capability filter

Table 8: Risks and fallbacks.

11 Positioning against Related Work

Verifying a language model's numbers against a trusted source is an active area. Proof-Carrying Numbers [14] requires each displayed number to link to a verified structured claim or be marked unverified. Tool receipts [15] check an agent's claims against signed records of the tool calls it actually made. Work on collective hallucination [16] decomposes multi-agent outputs into atomic claims. CricketMind v2 belongs to this family, and v3 must be presented against it. Three elements of v3 are, as far as a limited search found, not covered there:

1. **Subject binding.** Claims are tuples re-queried against the engine for their subject (C2), and the prose's named entities must agree with them (C6).
2. **Uncertainty faithfulness.** The wording of predictive and comparative claims is checked against an *external* engine's interval, one that includes estimation uncertainty (C4). Research on verbalised confidence [17, 18] studies how well a model states its *own* confidence, which is a different question.
3. **A static registry of claim types** that refuses, at build time, kinds of claims that no engine computes (I2).

In cricket, FanCric [19] coordinates language-model agents for fantasy team selection. CricBench [20] found that most SQL generated by language models for cricket analytics questions executes, while far less is semantically correct: direct evidence for the premise that cricket numbers produced by language models must be verified. Before any novelty claim is submitted, this positioning needs a systematic literature search.

12 What This Document Does Not Claim

- No v3 component has been built, and no v3 result has been measured.
- The worked examples use the synthetic corpus and say nothing about real players.
- The perception front end's accuracy is outside the scope of this document.
- The workload flag is not an injury predictor.
- Shot-keyed statistics on real data exist only if the commentary bridge passes validation.

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